

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2026

Regulation: R20 | University: JNTUK

Computer Networks (21CS52)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction, Switching & Physical Layer]

- a) Compare the OSI and TCP/IP reference models.
- b) Explain the different switching techniques with diagrams.

UNIT - 2 [Data Link Layer & MAC Protocols]

- a) Explain the working of CSMA/CD protocol.
- b) Explain error detection and correction using CRC and Hamming code.

UNIT - 3 [Network Layer, Addressing & Routing]

- a) Perform subnetting for the given IP address requirements.
- b) Explain distance vector and link state routing algorithms.

UNIT - 4 [Transport Layer & Congestion Control]

- a) Compare TCP and UDP with their header formats.
- b) Explain TCP congestion control mechanisms.

UNIT - 5 [Application Layer Protocols & Security]

- a) Explain the DNS hierarchy and name resolution process.
- b) Explain the DNS hierarchy and name resolution process.

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